

MFQ Juz 30: A Design-Science Framework for Mnemonic-Scaffolded Qur'anic Memorization Using Ordered Keywords, Semantic Mapping, Retrieval Practice, and Spaced Review

Methodological and Design-Science Article

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Manuscript status

This manuscript reports the design rationale, artifact architecture, and pre-empirical validation of MFQ Juz 30. It does not claim that the method improves memorization outcomes in learners. Effectiveness requires prospective human-participant testing with appropriate ethical approval, preregistration, and independent assessment.

Origin and development statement

The original concept and initial mnemonic system were conceived by Chilmar Buchari. The application architecture and detailed learning workflow were developed iteratively through structured dialogue between Chilmar Buchari and ChatGPT. Software code for the current prototype was generated by ChatGPT under human specification, review, correction, and acceptance. In accordance with current publication-ethics guidance, ChatGPT is not listed as an author because authorship requires human accountability.

Abstract

Memorizing the Qur'an requires high-fidelity sequential recall of a fixed verbal text, sustained revision, and resistance to order errors. Existing digital tools for Qur'anic memorization commonly emphasize repetition, recitation, testing, or monitoring, but less often formalize how a learner can move from a memorable cue structure to exact verse recall while progressively removing mnemonic support. This article presents MFQ Juz 30, a design-science framework and lightweight mobile-first prototype that combines a surah-level mnemonic anchor with bounded verse chunking, one ordered keyword per verse, short linked mnemonic stories, precise Arabic–Latin–Indonesian semantic mapping, verbatim Arabic triggers, retrieval practice, scaffold fading, and spaced review. The design was developed iteratively using a Design Science Research orientation and constrained by a religious-integrity baseline: mnemonic stories are memory aids rather than tafsir; the Qur'anic wording remains the terminal learning target; divine attributes and unseen realities are not represented literally; and representations of Prophet Muhammad are excluded. The current artifact covers all 37 surahs of Juz 30 (564 verses) organized into 127 mini-stories of no more than five verses each. Pre-empirical validation comprised structural checks, semantic mapping checks, designer-led mnemonic audits, religious guardrails, and software syntax/runtime checks. These procedures establish artifact completeness and internal consistency, not educational efficacy. The paper derives testable propositions and specifies a prospective randomized evaluation comparing MFQ-assisted learning with repetition-based and retrieval/spacing controls using delayed verbatim recall, verse-order accuracy, time-to-criterion, cognitive load, and retention as outcomes. MFQ is proposed as a falsifiable, auditable framework for investigating whether carefully faded mnemonic scaffolds can facilitate high-fidelity Qur'anic memorization without becoming a competing memory burden.

Keywords: Qur'an memorization; hifz; mnemonic; retrieval practice; spaced repetition; cognitive load; educational technology; design science; Juz 30

Highlights

- MFQ formalizes a cue-to-verbatim pathway rather than treating mnemonic imagery as the final learning target.
- The canonical unit is ≤ 5 verses, with one ordered keyword per verse and a 15–35-word linked mini-story.
- Every keyword is mapped to semantically aligned Arabic, transliteration, and Indonesian meaning fragments.
- Mnemonic support is explicitly faded toward unassisted recitation and scheduled delayed review.
- The present evidence is design/technical validation; learning effectiveness remains an empirical question.

1. Introduction

Qur'anic memorization (hifz) is an unusually demanding form of verbal learning. The learner is not merely required to remember gist, propositions, or concepts, but to reproduce a fixed sacred text in the correct sequence with high lexical fidelity. This makes hifz simultaneously a problem of encoding, ordered retrieval, error monitoring, and long-term maintenance. Traditional practice appropriately gives central importance to recitation, repetition (tikrar), teacher correction, and revision (muraja'ah). Digital systems have increasingly supplemented these practices through playback, testing, monitoring, speech recognition, and gamification. A literature review of Qur'an memorization technologies identified a growing ecosystem of applications and methods, while also showing substantial heterogeneity in features and pedagogical assumptions (Haryono et al., 2023).

The broader science of learning provides several mechanisms that are potentially relevant to hifz. Retrieval practice reliably strengthens long-term retention beyond additional study in many learning tasks (Roediger & Karpicke, 2006; Karpicke & Roediger, 2008; Karpicke & Blunt, 2011). Distributed practice and appropriately spaced review improve retention across delays (Cepeda et al., 2006, 2008). Mnemonic systems, including the method of loci and keyword methods, can improve ordered or paired-associate recall under suitable conditions (Twomey & Kroneisen, 2021; Qu et al., 2024). At the same time, cognitive-load theory warns that instructional aids can become counterproductive if they consume working-memory resources or create unnecessary integration demands (Paas & van Merriënboer, 2020). The design challenge is therefore not simply to add more cues, but to add the smallest useful cue and then remove it as learning stabilizes.

This challenge is particularly important in Qur'anic memorization. A mnemonic that is vivid but semantically loose may cause interference; a story that reverses verse order to become more entertaining can directly undermine the target sequence; and a visual aid that becomes the object of memorization can impose a second memory task. Moreover, religious content requires additional representational constraints: a learning aid must not be mistaken for tafsir, must not substitute for the Qur'anic wording, and should avoid inappropriate literal visualization of divine attributes, unseen realities, or the Prophet Muhammad. These requirements motivated the development of MFQ Juz 30 as a constrained mnemonic-scaffolding system rather than a free-form memory-story application.

The original Mnemonic Fonetik Qur'an (MFQ) concept was conceived by Chilmar Buchari as a systematic mnemonic framework. The Juz 30 project extends that surah-level anchoring logic to verse-sequence memorization. The current artifact integrates 37 surahs and 564 verses in a mobile-first application and formalizes a canonical pathway from an official surah anchor to short linked stories, verse-level keywords, semantic text mapping, exact Arabic triggers, retrieval, and fading. This paper

describes the design knowledge embodied in the artifact and, importantly, separates artifact validation from claims of learner effectiveness.

The study addresses three research questions: RQ1, how can a digital mnemonic framework support verse-order recall while preserving fidelity to the fixed Qur’anic text? RQ2, how can mnemonic scaffolds be designed to fade rather than become a competing target of memorization? RQ3, what technical, semantic, and religious-integrity gates are required before such a system is suitable for empirical evaluation?

2. Theoretical and Domain Foundations

2.1 Retrieval practice and durable retention

Testing is not only an assessment event; successful retrieval changes later accessibility of the retrieved material. Roediger and Karpicke (2006) demonstrated that testing can produce superior delayed retention compared with repeated study, and Karpicke and Roediger (2008) showed that repeated retrieval after successful recall is especially important for long-term learning. Retrieval practice has also outperformed elaborative concept mapping in some conditions (Karpicke & Blunt, 2011). For *hifz*, this literature implies that a digital tool should not remain a passive reading environment. It should repeatedly require the learner to generate the next verse or verse opening from a progressively weaker cue.

2.2 Spacing and revision schedules

Distributed practice is among the most robust learning effects in cognitive psychology. Cepeda et al. (2006) synthesized a large literature showing that temporal distribution affects verbal recall, and later work demonstrated that the optimal spacing gap depends on the desired retention interval (Cepeda et al., 2008). MFQ does not claim that one schedule is universally optimal; its current prototype uses a simple expanding sequence (1, 3, 7, 14, and 30 days) as an implementable review scaffold. Future adaptive scheduling should be driven by observed recall probability, verse difficulty, and user history rather than by a fixed calendar alone.

2.3 Mnemonics, ordered association, and keyword cues

Mnemonic methods can create additional retrieval routes by associating target material with memorable cues. A meta-analysis of the method of loci found a medium overall benefit across 13 randomized trials, while also noting limitations in experimental quality (Twomey & Kroneisen, 2021). Keyword mnemonics have a strong history in paired-associate learning, and recent work suggests that combining keyword mnemonics with constrained retrieval practice can improve vocabulary retention under some conditions (Qu et al., 2024). MFQ borrows the principle of memorable mediated association, but differs from a conventional phonological keyword method: its verse-level keyword is primarily a

semantic and sequential retrieval cue, and its terminal target is verbatim Arabic recitation rather than vocabulary translation.

2.4 Cognitive load, multimedia, and scaffold fading

Working memory is capacity-limited, making instructional economy central to mnemonic design. Cognitive-load theory emphasizes reducing extraneous processing and integrating necessary information so that working-memory resources can be used productively (Paas & van Merriënboer, 2020). The cognitive theory of multimedia learning similarly emphasizes limited capacity, dual channels, and active processing, and argues for coherent selection and integration of verbal and visual information (Mayer, 2024). These perspectives motivate three MFQ constraints: mini-stories are short; unnecessary objects and transitions are removed; and support is faded once a weaker cue is sufficient. In other words, vividness is subordinate to retrievability and cognitive economy.

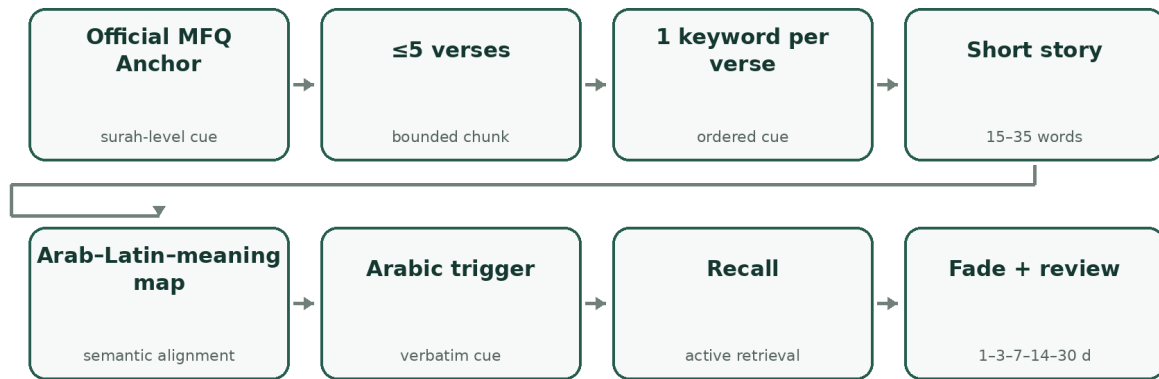
2.5 Qur’anic memorization as domain-specific expertise

Qur’anic memorization should not be equated with a general increase in memory capacity. Black et al. (2020) found no evidence that expert Qur’an memorizers had superior general verbal or visuospatial memory capacity compared with controls, illustrating the domain-specific nature of expertise. This finding is conceptually useful for MFQ: the system is designed to improve performance on a specific target task—accurate Qur’anic recall—not to claim far-transfer cognitive enhancement. Domain research also shows variation in learners’ preferred memorization approaches, including semantic, visual, and episodic tendencies (Muslim et al., 2024), supporting the value of testing MFQ across learner profiles rather than assuming uniform benefit.

2.6 Digital Qur’an memorization tools and the research gap

Digital Qur’an tools have explored repetition, mobile testing, mind maps, speech recognition, gamification, and monitoring. Haryono et al. (2023) reviewed 31 papers and 12 major memorization applications and documented the breadth of this design space. The present work addresses a narrower gap: a formally specified and auditable pathway that links a surah-level mnemonic anchor to ordered verse cues, exact-text triggers, retrieval, and fading while explicitly separating mnemonic assistance from the sacred source text. The contribution is therefore a design framework and artifact suitable for falsifiable evaluation, not a claim that digital mnemonic learning should replace teacher-guided hifz practice.

MFQ Juz 30 canonical learning pathway



Mnemonic scaffolding is designed to fade; the terminal target is unassisted Qur'anic recitation.

Figure 1. Canonical MFQ Juz 30 learning pathway.

3. Method: Design-Science Development

The work was organized as a design-science project because its primary research output is a purposeful learning artifact and a set of design rules intended to address a practical memorization problem. Design Science Research Methodology (DSRM) provides a suitable logic for problem identification, objective definition, design and development, demonstration, evaluation, and communication (Peppers et al., 2007). The present paper corresponds to an iterative build-and-evaluate phase and a pre-empirical evaluation stage. Human learning outcomes have not yet been tested.

3.1 Provenance of the concept and artifact

The initial MFQ concept, including the idea of stable mnemonic anchors for Qur'anic surahs, originated with Chilmar Buchari. The verse-level Juz 30 extension was developed through repeated design dialogues in which human requirements, corrections, religious constraints, and acceptance criteria were specified by Chilmar Buchari and candidate structures were generated, criticized, and revised with ChatGPT. ChatGPT generated the software code for the prototype under this iterative human direction. This provenance is reported because the role of generative AI is methodologically relevant to reproducibility and research integrity.

3.2 Design constraints

Design rule	Rationale
One verse, one primary keyword	Minimize cue ambiguity while preserving verse order.
≤5 verses per mini-story	Bound local memory load and preserve sequence manageability.

Design rule	Rationale
1–2 sentences; max 3; ~15–35 words	Prevent the mnemonic from becoming a second memorization task.
Story order = verse order	Sequence fidelity has priority over narrative elegance.
Keyword as event/action/object/sign	Convert an abstract list into an ordered imaginal sequence.
Arabic–Latin–meaning semantic map	Ensure each cue points to the same concept across representations.
Arabic trigger from the authentic verse	Transition from mnemonic cue to exact source wording.
Bridge between mini-stories	Reduce “restart” costs in long surahs without inventing tafsir.
Fade mnemonic support	Terminal target is unassisted Qur’anic recitation.
Religious guardrails	Avoid literal depiction of Allah, unseen realities, or Prophet Muhammad.

3.3 Canonical learning sequence

The canonical sequence is: Official MFQ Anchor → up to five verses → one keyword per verse → short linked story → precise Arabic–Latin–Indonesian keyword mapping → Arabic trigger → verse → bridge → recall → fade. The “official anchor” functions at surah level; the verse keyword functions at local sequence level. This separation is intended to reduce cross-surah interference while enabling verse-by-verse retrieval.

3.4 Three-layer semantic mapping

A central design refinement was the move from merely literal text matching to semantic mapping. Each keyword is linked to (a) the smallest useful Arabic fragment that carries the target concept, (b) the transliteration of the same Arabic fragment, and (c) the Indonesian translation fragment expressing the same concept. Negative keywords such as “not,” “without,” or “no” must include the negative operator in all three layers; highlighting only the positive noun is treated as pedagogically imprecise even when technically present in the verse. Whole-verse highlighting is permitted only as a temporary development fallback, not as a final mapping standard.

3.5 Religious-integrity guardrails

The system is intentionally non-tafsiric. Stories are explicitly labeled as memory aids and are not presented as explanations of revelation. The Qur’anic wording is the endpoint and may not be rewritten to fit a mnemonic. Attributes of Allah are represented, when needed, with text labels or abstract attentional cues rather than a body, location, or personification. Unseen realities such as angels, Paradise, Hell, the Throne, and eschatological events are treated as symbolic cues rather than claims about their true form. Prophet Muhammad is not visually depicted. These rules are not incidental content policies; they are part of the artifact’s design validity.

3.6 Iterative audit and correction

The artifact was repeatedly audited for six properties: story continuity, transition strength, visual distinctiveness, fidelity to verse order/meaning, memory load, and ability to cue the verse. A mini-story

passed the internal formative gate only when its mean audit score was at least 8.5/10 and no criterion was below 8.0; a score of 9.0 or higher was treated as a stronger freeze candidate. These ratings were designer-led formative judgments rather than independent psychometric measurements and therefore are not reported as evidence of effectiveness. They functioned as a structured defect-detection procedure during development.

4. The MFQ Juz 30 Artifact

4.1 Scope and content architecture

The current version covers all 37 surahs conventionally included in Juz 30, totaling 564 verses. Verses are organized into 127 mini-stories, each containing no more than five verses. Surahs are grouped by length into three working levels: short (Level 1), medium (Level 2), and longer stress-test surahs (Level 3). The level labels are implementation conveniences rather than claims about religious or linguistic difficulty.

Artifact element	Value	Status/interpretation
Juz 30 coverage	37 surahs	Complete
Verse coverage	564 verses	Complete
Mini-stories	127	≤5 verses each
Keyword mapping	564/564	Arabic–Latin–Indonesian semantic fragments
Public application name	MFQ Juz 30	Prototype version 1.7
Review schedule	1, 3, 7, 14, 30 days	Fixed prototype schedule; adaptive scheduling is future work

4.2 Scaffold stages

The application implements a staged reduction of help. Early stages expose the mnemonic story, keywords, source text, and semantic mapping. Intermediate stages retain keywords and Arabic triggers while reducing story dependence. Later stages emphasize retrieval from increasingly sparse prompts and eventually the verse alone. The design principle is “assistance should disappear as competence appears.” This aligns with guidance-fading logic in cognitive-load research, while acknowledging that the optimal fade schedule is an empirical question.

4.3 Mobile-first implementation

The artifact is implemented as a lightweight single-file web application optimized for smartphone use. The interface includes surah selection, official anchors, mini-story stations, verse displays, keyword highlights, guided recall, a mnemonic “movie” view, mastery stages, review queues, and local progress storage. Audio was deliberately excluded from the current prototype to keep the artifact lightweight and to avoid conflating the evaluation of mnemonic sequencing with the separate effects of auditory

modeling. A future efficacy study should add standardized expert-recited audio as a controlled factor rather than as an unmeasured feature.

4.4 Example of a mapping unit

For Al-Ikhlas 112:4, the experimental keyword “Not Equal” is linked to the Arabic fragment that includes the negation and equivalence phrase, its corresponding transliteration, and the Indonesian meaning “there is nothing equal to Him.” The design lesson is that mapping “equal” alone would be technically locatable but conceptually wrong because it drops the negation. This example illustrates why semantic validation is necessary in addition to substring validation.

5. Pre-Empirical Artifact Evaluation

The evaluation performed to date concerns artifact quality, completeness, and internal consistency. It should not be interpreted as evidence that MFQ improves memorization in learners. Five forms of pre-empirical evaluation were used.

Evaluation layer	What was checked	Current result
Structural	Coverage, ordering, chunk size, keyword count	37/37 surahs; 564/564 verses; 127 mini-stories; ≤5 verses per mini-story.
Text mapping	Literal and semantic correspondence	All 564 verse keywords mapped across Arabic, transliteration, and Indonesian meaning; final whole-verse fallbacks removed.
Mnemonic design	Continuity, transitions, visual stickiness, fidelity, load, verse cue	Iterative designer-led audit used to identify and repair weak chains and order errors.
Religious integrity	Non-tafsir framing, representation constraints	Guardrails embedded in story design, labels, and application notices.
Software	Data structure and code validity	Repeated syntax/runtime smoke checks during prototype iterations; no claim of device-wide usability validation.

One development example demonstrates the value of structured audit. In an early Al-‘Alaq chain, the narrative presented “feeling self-sufficient” before “transgressing,” whereas the intended verse order placed transgression before the self-sufficiency cue. The story sounded natural but violated the target sequence and was therefore redesigned. Similar audits found semantic-highlight problems in negative phrases where the positive noun was highlighted without the negation. These defects illustrate why mnemonic plausibility alone is insufficient; order and semantic precision must be machine- and human-checkable.

6. Proposed Cognitive Mechanism

MFQ is hypothesized to operate through a sequence of complementary mechanisms rather than through a single “memory trick.” First, the surah-level anchor provides a stable contextual cue. Second,

chunking limits the active sequence to a small number of verse cues. Third, one keyword per verse creates a sparse ordered index. Fourth, a brief linked story supplies transition structure and imagery. Fifth, semantic mapping connects the mnemonic cue back to authentic textual fragments. Sixth, Arabic triggers shift retrieval toward the exact source wording. Seventh, repeated retrieval and delayed review strengthen the target trace. Finally, fading attempts to prevent the scaffold from becoming retrieval-essential.

This mechanism yields a critical prediction: mnemonic support should be most beneficial during early sequence acquisition and should become unnecessary with successful retrieval. If learners remain dependent on the story at delayed tests, the design has failed its own objective even if immediate recall improves. Thus, MFQ should be evaluated not only on recall accuracy but also on cue dependence and the speed with which mnemonic support can be removed.

7. Testable Propositions

Proposition	Testable statement
P1	Compared with repetition-only study matched for exposure time, MFQ will reduce verse-order errors during early acquisition.
P2	MFQ combined with retrieval practice and spaced review will produce higher delayed verbatim recall than MFQ exposure without active retrieval.
P3	The benefit of MFQ will be mediated by reduced sequence uncertainty but may be offset when stories are rated as high in extraneous cognitive load.
P4	Semantic three-layer mapping will reduce cue-to-text mismatches compared with keyword labels that are not anchored to precise source fragments.
P5	A fading schedule will produce equal or better delayed recall with lower cue dependence than persistent full mnemonic support.
P6	Effects will vary with prior memorization, Arabic comprehension, imagery preference, and baseline retrieval ability; therefore, average treatment effects should be complemented by moderator analyses.

8. Proposed Empirical Evaluation Protocol

8.1 Study design

A strong next study would use a preregistered, assessor-blinded, randomized controlled design. Participants should be Muslim learners who can read the Arabic script at a predefined minimum level but have not already memorized the selected experimental surahs. Pre-screening is essential because Juz 30 familiarity is highly variable. Randomization should be stratified by baseline Arabic fluency, age group, and prior hifz experience.

A three-arm design would provide an interpretable decomposition of effects: (A) conventional repetition/restudy with the same Qur’anic text and study time; (B) retrieval practice plus the same spaced-review schedule but without MFQ stories; and (C) the full MFQ pathway with mnemonic anchors, ordered keywords, retrieval, fading, and spacing. This design allows the mnemonic contribution to be separated from the well-established effects of retrieval and spacing. A fourth arm using mnemonic support without retrieval could be added if sample size permits.

8.2 Outcomes

Class	Outcome	Operationalization
Primary 1	Delayed verbatim recall	Proportion of correctly recited words/verses at 7 and 30 days; scored by blinded qualified assessors.
Primary 2	Verse-order accuracy	Omissions, transpositions, anticipations, and incorrect next-verse transitions.
Primary 3	Time-to-criterion	Minutes/trials required to reach a prespecified accurate-recitation criterion.
Secondary	Immediate recall	Post-learning accuracy after standardized study exposure.
Secondary	Cue dependence	Recall after story/keyword cues are removed; number of rescue cues required.
Secondary	Cognitive load	Validated mental-effort/cognitive-load rating after each learning block.
Secondary	Usability and acceptability	System Usability Scale or equivalent plus domain-specific acceptability questions.
Exploratory	Retention trajectory	Repeated measures across 1, 3, 7, 14, and 30 days.

8.3 Statistical analysis

The primary analysis should be intention-to-treat. Delayed recall can be modeled with mixed-effects regression including participant and surah as random effects, fixed effects for study arm, time, and their interaction, and prespecified covariates for baseline Arabic fluency and prior memorization experience. Order errors can be modeled as counts or binary transition-level outcomes. Time-to-criterion can be analyzed using survival or accelerated-failure-time methods if censoring occurs. The sample size should be determined prospectively from a smallest effect of interest rather than from a conventional “medium effect” assumption. The analysis plan, exclusion rules, outcome hierarchy, and stopping rules should be preregistered.

8.4 Fidelity and religious review

Educational efficacy should not be evaluated independently of textual and religious fidelity. The final research version should undergo verse-by-verse verification against the selected official Qur’anic source, review by qualified Qur’an teachers for recitation and pedagogical appropriateness, and review of

mnemonic scenes for religious acceptability. Any mnemonic that improves recall but introduces a systematic wording, sequence, or conceptual error should be considered a design failure rather than a successful trade-off.

9. Discussion

MFQ Juz 30 contributes a design proposition to the intersection of learning science, religious education, and educational technology: a mnemonic can be useful precisely because it is temporary. Many memory aids are evaluated by how memorable the aid itself becomes. In high-fidelity text memorization, however, the more relevant criterion is whether the aid accelerates access to the target and then becomes unnecessary. This “designed disappearance” is the organizing principle of MFQ.

The framework also introduces auditable semantics into mnemonic design. A keyword is not accepted merely because it feels related to a verse; it must be traceable to aligned Arabic, transliteration, and meaning fragments. This creates an explicit bridge between imaginative memory support and textual fidelity. The requirement is particularly important for negation, contrast, and multi-part concepts, where a shortened highlight can reverse or dilute meaning. Such auditable mappings may be useful beyond Qur’anic memorization in any learning domain where mnemonic flexibility must coexist with a fixed authoritative source.

A second contribution is methodological transparency around religious representation. Conventional multimedia design theory focuses mainly on cognitive effectiveness. MFQ adds domain-integrity constraints that are treated as first-class design requirements rather than post hoc content moderation. This is important because a visual or narrative cue can be cognitively effective yet inappropriate for the religious domain. The framework therefore evaluates utility within a constrained solution space.

A third contribution is the explicit separation of design validation from efficacy validation. The artifact is complete, internally consistent, and technically testable, but no participant data currently demonstrate superior learning. This distinction is especially important when generative AI has contributed to design and coding, because fluent software and polished narratives can create a misleading impression of scientific validation. The next scientific milestone is therefore not another interface refinement but a controlled human study with independent assessors and preregistered outcomes.

If empirical support is obtained, several extensions become plausible: adaptive spacing based on recall probability, personalized selection of mnemonic modality, audio-guided pronunciation, teacher dashboards, and controlled comparisons of self-generated versus system-provided mnemonics. If the method fails to outperform retrieval-and-spacing controls, that result would still be theoretically

informative by showing that additional mnemonic mediation adds little or imposes avoidable load in this domain. The framework is deliberately falsifiable.

10. Limitations

- The current evidence is pre-empirical. No human participant has yet been randomized to MFQ versus a comparator in the present project.
- The mnemonic audit was primarily designer-led and was not conducted by independent raters; inter-rater reliability is therefore unavailable.
- The fixed 1–3–7–14–30-day review schedule is a prototype heuristic rather than an optimized spacing algorithm.
- The application currently emphasizes sequence and recall scaffolding and does not independently validate tajwid, pronunciation, or teacher-led correction.
- Visual imagery may benefit some learners more than others; semantic or auditory learners may require different cue formats.
- Juz 30 contains familiar surahs for many Muslims, creating prior-knowledge and ceiling-effect risks in future experiments.
- Indonesian translation mapping is useful for Indonesian learners but may not generalize directly across languages or translation traditions.
- The use of generative AI in iterative design and programming requires unusually explicit provenance, verification, and reproducibility documentation.

11. Implications for Research and Practice

For researchers, MFQ provides a defined intervention whose components can be experimentally dismantled: surah anchors, verse keywords, story linkage, semantic mapping, Arabic triggers, retrieval, spacing, and fading. This supports component-level causal questions rather than treating “an app” as a monolithic treatment. For Qur’an educators, the framework should be viewed as a supplementary scaffold, not a replacement for qualified instruction, listening, correction, or muraja’ah. Its strongest potential role may be the early stabilization of verse sequence and the organization of independent review between teacher sessions.

12. Conclusion

MFQ Juz 30 is a complete design-science artifact that operationalizes a constrained mnemonic approach to Qur’anic memorization across 37 surahs and 564 verses. Its core innovation is not the use of stories alone, but the integration of stable surah anchors, bounded verse chunks, one ordered keyword per verse, short linked stories, precise Arabic–Latin–meaning mapping, authentic Arabic triggers, active

retrieval, delayed review, and deliberate fading. The artifact has passed extensive internal structural, semantic, religious-integrity, and software checks, but it has not yet established educational effectiveness. The scientifically appropriate next step is a preregistered controlled trial that measures delayed verbatim recall, verse-order errors, time-to-criterion, cognitive load, cue dependence, and retention. Such a study would determine whether temporary mnemonic scaffolding can improve the acquisition and maintenance of high-fidelity sacred-text recall beyond the benefits of retrieval practice and spacing alone.

Declarations

Ethics statement

No human participants, identifiable personal data, or biological materials were involved in the design and pre-empirical artifact validation reported in this manuscript. Prospective learner evaluation will require approval from an appropriate ethics committee/institutional review board and informed consent (and parental consent/assent where minors are involved).

Data and code availability

The prototype artifact, design baseline, and versioned mapping data can be archived in a persistent repository at the time of submission. A public release should follow final source-text validation and removal of any development-only metadata. A DOI-bearing repository (e.g., Zenodo/OSF) is recommended for reproducibility.

Conflict of interest

The author declares no competing interest related to the scientific evaluation proposed in this manuscript. Any future commercialization, licensing, or institutional deployment of MFQ should be disclosed in the submitted version.

Funding

No external funding is reported for the present design-development stage. [Update if funding is obtained before submission.]

Author contributions (CRediT-style)

Chilmar Buchari: Conceptualization; original mnemonic framework; methodology; requirements specification; domain constraints; iterative validation; project administration; supervision of application development; review and editing; final accountability for the work.

Declaration of generative AI and AI-assisted technologies

ChatGPT (OpenAI) was used as an AI-assisted design and software-development tool. The original MFQ idea and initial mnemonic concept were created by Chilmar Buchari. The detailed application design was developed through iterative dialogue between Chilmar Buchari and ChatGPT. ChatGPT generated the application code (programming) and assisted with drafting, restructuring, and literature-oriented manuscript preparation. Chilmar Buchari provided the design objectives, corrections, acceptance/rejection decisions, religious constraints, and final human responsibility. ChatGPT is not an author and cannot take responsibility for the accuracy, integrity, or originality of the work. All AI-generated content must be reviewed, source-checked, and approved by the human author prior to submission.

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Appendix A. Submission-Readiness Checklist for a High-Impact Journal

- Confirm final author affiliation, correspondence email, and ORCID.
- Archive code, design baseline, version history, and a reproducible mapping dataset in a DOI-bearing repository.
- Obtain independent Qur'an-text verification and domain-expert religious/pedagogical review.
- Convert the present design paper into an empirical article once a preregistered controlled study is completed.
- Report CONSORT-style participant flow if randomized; provide a preregistration link and analysis code.
- Replace internal formative scores with blinded assessor outcomes, inter-rater reliability, confidence intervals, and effect sizes.
- Include a journal-specific generative-AI declaration and follow the target journal's current policy at submission time.